CURRICULUM VITAE

BADAM PRAVEENKUMAR

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Personal Profile / Personal Attributes :

- Commitment to the Highest level of Professional and Personal Excellence, Good team Leader and Lead the team for the Technical aspects and Project deliverables
- Seeking new responsibilities and takes initiative; Self-sufficient and proved to be very efficient in working under pressure
- Good interpersonal and communication skills - Works well with others and communicates, Motivates and Encourages
- Extremely reliable and dependable - Analytical and Questioning, Strives for quality
- Strategic, Proactive and Innovative- Anticipates and takes initiative

Experience:

- 20+Yrs of experience in Design, Development and Testing of RF/Microwave Component Level and System level.
- Experience in the RADAR, Navigation Related technologies for Avionics applications and Wireless technologies for Industrial, Home and Building applications,WLAN, WCDMA, and LTE Technologies, Computer based Train Control (CBTC) and Multimedia (MM) application for Kochi Metro Train.
- Experience in Radio board design, Development and Testing experience to implement large scale end-end communication projects
- Cascaded Analysis for the Radio Systems ,Link budget analysis for Indoor Wireless Products and Kochi Metro Train
- Collaboratively working across different domain teams while executing Aerospace ,BTS and Wireless projects
- Experience in Wireless technologies like Zigbee, WLAN transceiver design and testing for Industrial, Home control applications, WCDMA and LTEfor Macro BTS application
- Evaluation of Clemente Boards for NEC LTE Macro BTS Application
- Experience in RF Designing Software like Advanced Design System (ADS) of HP Agilent and EESOF, AWR MWO, Visual System Simulator (VSS), Linecalc, Appcad, ANSYS HFSS, MATLAB.
- Experience in design of RF/Microwave Components like Filters, Power Dividers, Directional Couplers, Low Noise Amplifiers, PLL, ADC, DAC and Automatic Gain Control (AGC) System.
- Experience in PCB Layout simulation and Co-Simualtion
- Experience in PCB Antenna design and Antenna Performance and System Performance characterisation in Anechoic chamber
- Experience in Designing of Receiver for Tactical Air Navigation System (TACAN)
- Experience in Installation and testing of IFF1410 A for Jaguar Air Craft
- Experience in System Level Simulation of a Receiver and Transmitters using AWR VSS
- Experience in Layout review using Cadence Allegro viewer
- Extensive knowledge of PCB designing considering EMI/EMC aspects
- Expertise in using Cadence Allegro and Mentor Expedition and DX-Designer
- Experience in Integration of other Line Replacement Units (LRU’s)
- Experience in Tx and Rx line-up analysis, Component simulations, System requirement Analysis and System simulations
- Exposure in Avionics Procedures likes Vibration, EMI/EMC, CATH, Shock and Acceleration.
- Hands on experience in Several RF instruments like HP Phase noise meter, Spectrum Analysers, Network analysers, Noise figure meter, One Wireless Tester, Signal Generators, and CRO.
- Managing the team for Technical Aspects and Lab establishment during project start-up

Career History:

- Working as **Senior System Architect** in Unistring Tech Solutions PVT., LTD Hyderabad
- Aug 2018- Dec’2021 worked as **Senior Manager** in Astra Microwave Products Limited (AMPL),Hyderabad
- Jan’2017 – Aug’2018 Worked as **RF and Networking Engineer** in Alstom Transport, Bangalore
- Sep’2011 – Jan’2017 Worked as **Lead Consultant** in Wipro Technology Solutions, Bangalore.
- Feb’2006 – Sep’2011: Worked as **Senior Engineer (Design)** in Honeywell Technology Solutions, Bangalore.
- Jan’2009 – Apr’2009: Worked with Honeywell International Aerospace division Redmond-USA for
- RDR-4000 Product Maintenance and Improvement (PMI) redesign activities and in Olathe trained on test
- Setup and testing individual CCA’s of Weather RADAR.

- Jan’2004 – Feb’2006: Worked as Asst. Engineer (Design) in SLRDC, with **“Hindustan Aeronautics Ltd**”, (HAL) Hyderabad. SLRDC is a **“STARTEGIC ELECTROINCS RECEARCH AND DESIGN CENTER** “Premier R&D Center in India for Avionics.

Educational Qualifications:

- M.Tech (ECE)(Microwave & Radar Engineering) with71% from College of Engineering, Osmania University, Hyderabad, A.P.
- Achieved GATE score of 94.63% in Feb’2002
- Bachelor of Engineering (ECE) with 71% from College of Engineering, Osmania University, Hyderabad, A.P.
- Diploma (ECE) with 83% from Govt. Polytechnic, SBTET, Warangal, A.P.

Professional Awards:

- **“Individual Excellence Award”** – Honeywell highest award in recognition of remarkable performance, excellent contribution and dedicated performance for the design and validation during One Wireless program
- **“Honeywell Special Award”** –in recognition of remarkable performance, excellent contribution and dedicated performance for design and validation during One Wireless program
- **‘’Technical/Innovation Excellence Award”** in recognition for outstanding technical/innovation achievements in Project “ One Wireless (RAPTOR)” for the year 2007
- **“Team Achievement Award”** for outstanding performance and excellent contribution towards RAPTOR Wireless Testing
- **“Partner Award”** for excellence contribution towards successful and high quality release of One Wireless Program (R110) through focused, dedicated and diligent team work. This is a major mile stone for Honeywell ACS HPS business
- Programming with IOT Boards certificate offered by “Pohang university”

Achievements:

- Participated training programme on “Atmosphere science and Remote Sensing techniques” conducted by **“NATIONAL MST RADAR FACILITY (NMRF)”** Department of space, Gadanki, Tirupati – A.P.
- Trained on Management and personality development at **“Hindustan Management Academy (HMA)”,** Bangalore.
- Managed to deliver consistently high quality deliverables and services to client in all situations.
- Trained on green belt and implemented Six Sigma processes

Technical Skills:

- **Hardware Platforms:**RF Designing, Analog and Mixed Signal
- **Operating Systems :**Windows XP, Windows 95,Windows NT, UNIX, DOSLanguage,C, C++, Assembly language of 8085,Microwave Office, Advanced Design Systems, EESOF,Linecalc,Pspice,
- **PCB Designing tools**: ORCAD, AutoCAD, Cadence Allegro
- **Domains:**RF, Wireless, RADAR, Analog, Aero HW
- **Programming skills**: C, TI Smart RF studio, IAR Embedded work bench, Verilog, Labview for Automation, able to program Transceiver IC’s
- **RF simulation tools**: Advanced Design System (ADS),SystemVue, AWR Microwave Office and VSS, Ansys HFSS

Projects Profile:

1. Project Name: S-Band RADAR Target Simulator (RTS)

Organization: Unistring Tech Solutions PVT., LTD
Tools: Advanced Design System (ADS), SystemVue, HFSS
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 4
Duration: June’2022 to Mar’2023
Description: Radar Target Simulator (RTS) is used to test and quantify the parameters of radar by emulating a real environment the radar encounters like different types of targets, different types of clutter at different ranges and speeds. The function of RTS in radiation mode is to intercept the transmitted energy, down convert it and modify the transmitted signal to generate targets of different cross sections, ranges and Doppler’s and retransmit it towards multiple receive stations. All the receive stations and the transmit station may be placed around 10 Km from RTS tower which is around 50 ft. high. In addition, the RTS parameters should be programmable through a high speed Ethernet interface from RTS station.

Contribution:

- Preparation of High Level Architecture Block diagram , Individual component simulation, Cascaded System budget simulations for Tx and Rx
- Time domain Analysis in System Vue for Spectrum O/p Power(dBm), Phase Noise(dBc) , Spurs and Harmonics levels(dB)
- Finalisation of Architecture components, ADRV9009 Transceiver evaluation
- Individual LRU's Design and testing ,Front End Band Pass Filter design and fabrication for the S- Band (ie., 2.89-3.31 GHz) in ADS and Measured fabricated BPF Performance
- Schematic Capture and Layout Preparation
- EM Co-simulation and Layout Parametric Optimisation
- Mechanical Enclosure finalisation, fixing of Individual LRU's and Routing

2. Project Name: AD RADAR RX System Performance Simulation and Optimisation

Organization: Unistring Tech Solutions PVT., LTD
Tools: Advanced Design System (ADS), SystemVue, HFSS
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 1
Duration: Mar'2022 to May'2022
Description: ADRADAR Rx System Operates within the frequency band of 9.35GHz-9.45GHz and down convert to IF frequency of 70MHz+/-7.5MHz. Band Reject filter is chosen to meet the dynamic range requirement of -14dBm to-127dBm and the System not have the external controls to adjust the Gain, Mixer Two tone analysis for verification of Spurious and Harmonic

Contribution:

- Preparation of High Level Architecture Block diagram , Individual component simulation, Cascaded System budget simulations for Rx to verify the requirement of Cascaded Gain(dB),Noise Figure(dB) and Minimum Detectable Sensivity(MDS)(dBm)
- Time domain Analysis(**ie.,**Data flow analysis) in System Vue for checking the Spectrum O/p Power(dBm), Phase Noise(dBc) , Spurs and Harmonics levels(dB), and Doppler freq.Shift(KHz),

3. Project Name: 30W Power Amplifier design for X-band

Organization: Unistring Tech Solutions PVT., LTD
Tools: Advanced Design System (ADS), SystemVue, HFSS
HW Instruments: Spectrum Analyser, Signal Generator, Power Meter
Team Size: 4
Duration: Aug'2022 to Feb'2023
Description:

- Radar Target Simulator is used to test and quantify the parameters of Radar by emulating a real environment the radar encounters like different types of targets, different types of clutter at different ranges and speeds. The function of RTS in radiation mode is to intercept the transmitted energy, down convert it and modify the transmitted signal to generate targets of different cross sections, ranges and Doppler's and retransmit it towards multiple receive stations. All the receive stations and the transmit station may be placed around 10 Km from RTS tower which is around 50 ft. high. In addition, the RTS parameters should be programmable through a high speed Ethernet interface from RTS station.
- 30W-PA Module performs amplifying RF signal in X-band (**ie.,** 9.2 to 9.8GHz) to generate 30W output power (dBm) at defined maximum DC power consumption, HW interfaces are RF input, RF output, Power Supply connections, Sequential Biasing Gate control (Vg) and Drain Control (Vd) for driver and Final Power Amplifier IC. Additional Potential divider is implemented in Sequential biasing circuit Gate (Vg) Control Voltage .It Consists of Switch , Driver Amplifier, Final PA and Isolator

Contribution:

- Preparation of High Level Architecture Block diagram, Individual component simulation, Cascaded System budget simulations.
- Cascaded Analysis in System Vue for O/p Power (dBm), Cascaded Gain (dB), OP3dB (dBm)
- Sequential Biasing Ckt design and Carrying Power Budget
- Schematic Capture and Layout Preparation
- Individual LRU's design (**ie.,** SPDT SW, Driver Amplifier, Power Amplifier, and Isolator) and Characterising the Performance.
- EM Co-simulation and Layout Parametric Optimisation
- Mechanical enclosure finalisation, Placement of Individual LRU's, Power Supply Module, i/p and O/p Connector and Routing in enclosure
- Resonant Mode Analysis using *.step file in ANSYS

- System Level testing for O/p Power(dBm) ,Gain(dB) ,Power Added Efficiency (PAE) ,Harmonics and Spurious levels(dB) at Ambient and extreme Operating Temp.conditions(-40^{0}C to +55^{0}C)

4. Project Name: 35W Wideband Power Amplifier design

Organization: Unistring Tech Solutions PVT., LTD
Tools: Advanced Design System (ADS), SystemVue, HFSS
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 4
Duration: Nov'2022 to Aug'2023
Description:

- ➢ Radar Target Simulator (RTS) is used to test and quantify the parameters of radar by emulating a real environment the radar encounters like different types of targets, different types of clutter at different ranges and speeds. The function of RTS in radiation mode is to intercept the transmitted energy, down convert it and modify the transmitted signal to generate targets of different cross sections, ranges and Doppler's and retransmit it towards multiple receive stations. All the receive stations and the transmit station may be placed around 10Km from RTS tower which is around 50 ft.high. In addition, the RTS parameters should be programmable through a high speed Ethernet interface from RTS station.

- ➢ 35W-Wideband PA Module performs amplifying RF signal (**ie.,** 0.5GHz to 6.0GHz) to 35W output power level at defined maximum DC power consumption, HW interfaces are RF input, RF output, Power Supply connections, Sequential Biasing Gate control (Vg) and Drain Control (Vd) for driver and Final Power Amplifier IC. Additional Potential divider is implemented in Sequential biasing circuit Gate (Vg) Control Voltage. It Consists of Switch , Predriver, Driver Amplifier, Final PA and Isolator

Contribution:

- Preparation of High Level Architecture Block diagram, Individual component simulation, Cascaded System budget simulations for PA system.
- Cascaded Analysis in System Vue for O/p Power (dBm), Cascaded Gain (dB), OP3dB (dBm) and Sequential Biasing ckt design and Carrying Power Budget
- Schematic Capture and Layout Preparation
- Individual LRU's design (**ie.,** SPDT SW, Predriver, Driver Amplifier, Power Amplifier, and Isolator) and characterising the Performance.
- EM Co-simulation and Layout Parametric Optimisation
- Mechanical Enclosure finalisation, Resonant Mode Analysis using (*.step) file in ANSYS for fixing of Individual LRU's, Power Supply Module, i/p and O/p Connector and Routing
- System Level testing for O/p Power(dBm) , Gain(dB) ,Power Added Efficiency (PAE) ,Harmonics and Spurious levels(dB) at Ambient and extreme Operating Temp.conditions(-40^0 to +55^{0}C)

5. Project Name: 5.2 GHz to 5.9GHz 200W Power Amplifier design

Organization: Unistring Tech Solutions PVT., LTD
Tools: Advanced Design System (ADS), SystemVue, HFSS
HW Instruments: Spectrum Analyser, Signal Generator, Power Meter
Team Size: 4
Duration: Oct'2023 to Mar'2024
Description:

- ➢ Radar Target Simulator is used to test and quantify the parameters of Radar by emulating a real environment the radar encounters like different types of targets, different types of clutter at different ranges and speeds. The function of RTS in radiation mode is to intercept the transmitted energy, down convert it and modify the transmitted signal to generate targets of different cross sections, ranges and Doppler's and retransmit it towards multiple receive stations. All the receive stations and the transmit station may be placed around 10 Km from RTS tower which is around 50 ft. high. In addition, the RTS parameters should be programmable through a high speed Ethernet interface from RTS station.
- ➢ 200W-PA Module Consists of two individual 100W Chains for amplifying RF signal from 5.2 to 5.9GHz to generate 200W output power (dBm) at defined maximum DC power consumption; HW interfaces are RF input, RF output, Power Supply connections, Sequential Biasing Gate control (Vg) and Drain Control (Vd) for driver and Final Power Amplifier IC. Additional Potential divider is implemented in Sequential biasing circuit Gate (Vg) Control Voltage. Complete 200W PA Chain Consists individual 100W PA chain of Predriver Amplifier, Driver Amplifier, Final PA , Isolator and High Power combiner using Rat Race CPLR

Contribution:

- Preparation of High Level Architecture Block diagram, Individual component simulation, Cascaded System budget simulations.
- Cascaded Analysis in System Vue for O/p Power (dBm), Cascaded Gain (dB), OP3dB (dBm)
- Sequential Biasing Ckt design and Carrying Power Budget

- Schematic Capture and Layout Preparation
- Individual LRU's design (**ie.,**1:2 Power divider, Pre-Driver Amplifier, Driver Amplifier, Power Amplifier, Isolator and Rat Race CPLR) and Characterising the Performance.
- EM Co-simulation and Layout Parametric Optimisation
- Mechanical enclosure finalisation, Placement of Individual LRU's, Power Supply Module, i/p and O/p Connector and Routing in enclosure
- Resonant Mode Analysis using *.step file in ANSYS
- System Level testing for O/p Power(dBm) , Gain(dB) ,Power Added Efficiency(PAE) ,Harmonics and Spurious levels(dB) at Ambient and extreme Operating Temp.conditions(-40^{0}C to +55^{0}C)

6. Project Name: X-Band Seeker for Active Electronic Scanning Array (AESA)

Organization: Astra Microwave Products Limited (AMPL)
Tools: Advanced Design System (ADS), SystemVue
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 2
Duration: June'2019 to Nov'2019
Description: Active Electronic Scanning Array (AESA) System receives an X-band RF Signal which is distributed among the TR Module planks for the amplification. Every Module Phase shifts the phase of signal as required by the plank controller and amplifies the RF Signal to 4W.Theamplified signal is fed to antenna and Radiated. The Received Echo is routed through the receiver portion of TR module and after amplification the outputs of 144 Elements are added in quadrant wise to generate 4 signals, these signals are fed to the Monopulse comparator which generates the Sum Signal, Azimuth Difference Signal, Elevation difference signal.

Contribution:
- Design of the **1:2,1:3,1:4,1:5,1:6**,**1:7**and **1:10** Equal and Unequal Power divider simulation for power distribution to the Individual TRM Module
- Tx and RX system design with ASL6001OPGaAs MMIC Core Chip
- Individual component simulation, Cascaded System budget simulation of Tx and Rx
- Integration of Plank controller and Array Controller with TRM Module
- Schematic Capture and Layout Preparation
- EM Co-simulation and Layout Parametric Optimisation

7. Project Name: Ku-Band Seeker Rx, Exciter for Active Electronic Scanning Array(AESA)

Organization: Astra Microwave Products Limited (AMPL)
Tools: Advanced Design System (ADS), SystemVue
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 2
Duration: Jul'2021 to Oct'2021
Description:
Active Electronic Scanning Array (AESA) System receives a Ku-band RF Signal which is distributed among the TR Module Planks for the amplification. Every TR Module Phase shifts the phase of signal as required by the plank controller and amplifies the RF Signal to 4W.The amplified signal is fed to antenna and Radiated. The Received Echo is routed through the receiver portion of TR module and after amplification the outputs of 144 Elements are added in quadrant wise to generate four Signals, these signals are fed to the Monopulse Comparator which generates the Sum Signal, Azimuth Difference Signal, Elevation difference signal. The Ku-Band Exciter generate STALO-1,-2, TX drive O/p, Pilot Signal O/p, Test signal O/p

Contribution:
- Rx Architecture finalisation and component selection, Cascaded System Analysis in Systemvue to verify the System parameters **ie**., Cascaded Gain, Noise Figure, Input P1dB (dBm), IIP3(dBm)
- Schematic Capture, Layout generation and PDR document preparation, Mixer Spur Analysis and selection of LO frequency, Phase Noise analysis at LO,Tx ports

8. Project Name: Wideband TRM Module Design and Development

Organization: Astra Microwave Products Limited (AMPL)
Tools: Advanced Design System (ADS), SystemVue
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 2
Duration: Sep'2019 to Mar'2020
Description:
Wideband TRM module operates in the frequency range of 1-6GHz which contains dual Transmit and Receive Chains, each transmit chain delivers o/p power of 10W based on chain selection and each Rx chain is capable to

provide gain of 30dB for the min i/p signal level of -70dBm. The common path for both Tx and Rx contains quad Phase shifter to cover the operating frequency band and also the Digital Control attenuator, SP4T SW, Gain amplifier

Contribution:

- Cascaded System Budget Analysis w.r.t Pout, Gain, Noise Figure using ADS Systemvue
- Individual Component Simulations and tuning to meet the System Specification
- Power Supply Board design for the RF and FPGA board and DC Power Analysis
- Schematic Capture and Layout making
- EM layout simulation using ADS

9. Project Name: SAAB FAM project

Organization: Astra Microwave Products Limited (AMPL)
Tools: Advanced Design System (ADS)
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 2
Duration: Nov'2019 to Jan'2020
Description:
Wideband TRM module operates in the frequency range of 1-6GHz which contains dual Transmit and Receive Chains each transmit chain delivers o/p power of 10W based on chain selection and each Rx chain is capable to provide gain of 30dB for the min i/p signal level of -70dBm. The common path for both Tx and Rx contains quad Phase shifter to cover the operating frequency band and also the Digital Control attenuator, SP4T SW, Gain amplifier

Contribution:

- Design of Master Power Supply(MAPS) and Local Power Supply (LPS)Board
- Individual Component Simulations and tuning to meet the System Specification
- Power Supply Board design for the RF and FPGA board and DC Power Analysis
- Schematic capture and Layout preparation

10. Project Name: MOXA based Multimedia CCTV solution designed for Kochi Metro Project (KMP)

Organization: Alstom India Ltd
Networking Tools: Wireshark Capture tool, LAN breaker tool, LTFMT, RCLP, Ixia Chariot Console
HW Instruments: Spectrum Analyser, Signal Generator
Team Size: 2
Duration: Feb'2017 to Aug' 2018
Description:
The DCS Signalling system consists of a Radio system used to exchange control information between Train and Control centre. The DCS Multimedia-CCTV system consists of a radio system, used to transmit multimedia-CCTV traffic from on-board equipment to OCC/BCC through trackside equipment. Both the Radio systems shall provide high quality communication services. The DCS Radio system is composed of a set of Access points spread along the track connected to directive antennas and Radio modems installed inside each train also connected to directive antennas. MOXA wireless solution was selected based on PoC tests and architecture as required by the contract The TAPs are sequentially interconnected in series using proprietary MOXA Turbo Chain topology (Daisy Chain architecture); with the end devices connected to respective L2 switches at the stations. The traffic will follow always towards the primary direction and will be diverted to secondary direction at the time of failure on the L2 switch (or) fiber cut. All the L2 switched are connected by MOXA TURBO ring topology and works on RSTP for link recovery. The L2 switches (non-redundant) are connected to redundant L3 switches, the wayside access controllers (WAC) are connected in redundant configuration to main and redundant L3 switches at OCC ,All the roaming decisions in Moxa Multimedia are centralized i.e. Controller based Roaming. To overcome slow handoffs on high speed railways, Moxa's rail specific Turbo Roaming technology uses WAC (wireless access controllers) that provides centralized security management. When roaming to the next available AP, the client can be pre-authenticated by the WAC, which eliminates the need for authentication, and reduces the handoff time to 50ms

Contribution:

- Evaluation of 2.4GHz Moxa Wireless Radios Performance for Sensitivity, ACPR ,Interference and Blocking Performance
- Finalisation of WAC parameters for Roaming decision
- Capturing of Syslog parameters (***.pcap)**file using Wireshark capture tool and Plotting with RSSI transformer tool
- Calculation of Handoff Parameters using LAN breaker tool and verifying whether the Handoff is happening as per the priority settings in WAC

- Preparation of Moxa Technical document, SIT-19 Radio Coverage Document, SIT-20 End-End Communication test procedure document for Multimedia and Test report PPT's
- Interference Survey and establishment of the reliable Network Plan

11. Project Name: ADF703X (ADI) Antenna Diversity Daughter Board development and Firmware Characterisation

Organization: Wipro Technology
Client: Analog Device (ADI)
RF tools: ADS, LabVIEW
Team Size: 2
Duration: Jun'2015 to Nov'2016
Description:
ADF7033 is a low Power Integrated Radio Transceiver that supports wide range of modulation schemes (**ie., 2(G)FSK, 4(G)FSK, OOK, GMSK**), frequency bands and Channel BW's., ADF7030 Antenna Diversity Daughter board is designed to improve the performance of the radio link, Antenna Switch component select one of two spatially separated antennas to be connected to the receiver input depending on the logic levels applied to the Antenna Switch selection pins from the ADF7033. GPIO's used to drive control voltage (**VC1**&**VC2**) of external antenna Switches. **ADF7033** switch between the two antennas while searching for a valid Preamble sequence and at the same time measuring the RSSI.

Contribution:
- Requirements capture, selection of Front End switch components
- Design of Rx Front end Matching ckt and Tx Harmonic filter using *.S2P model
- Circuit and Layout simulation using RF design tool (ADS)
- BOM capture and following up getting of components
- Schematic review using Cadence Allegro 16.5 Concept HDLand layout review in Cadence Allegro PCB Librarian XL tool
- Characterisation of ADF7030 RX and TX test cases for different Firmware patches using Labview VI and test report PPT preparation
- Customer and Vendor interactions for lab establishment

12. Project Name: All- In- One eNB for LTE-A Macro BTS

Client: NEC
Organization: Wipro Technology
RF tools: AWR Microwave Office, Visual System VSS, SystemVue, ADS, HFSS
Team Size: 4
Duration: Aug'2013 to Jun'2015
Description: All in one (AIO) BTS is a single unit LTE-A Macro BTS with 5W total output power. AIO BTS HW platform will be the 3GPPrelease 9 compliant product in the first release and it will support LTE-advance feature with SW upgrade. AIO BTS product will be designed to provide greater capacity in the proposed product segment and shall provide greater deployment flexibility. Initial design is for the Band-41 and later needs to support for TDD Band-39,-40,-42,-43and LTE FDD Operation from 700MHz-800MHz Band with minimal front end components change without any layout changes

Contribution:
- Involved in HW Requirement study, RF requirement Feasibility analysis for Band-41
- Line-up calculations for Tx and RX using Clemente Transceiver IC,front end components, PA and BPF
- Selection of Front end Component like Balun,SAW Filter for Tx and Rx, Simulation of components and Matching Ckt design with Clemente Tx and Rx ***.S2P** Model,
- Cavity Mode Resonant analysis with RF Cover, Case and Cavity using HFSS
- Thermal Analysis of Tx and Rx for calculation of heat sink requirement,Power Supply sequence
- Evaluation Clemente Tx and Rx boards ,Spurious& Harmonic analysis for Tx and Blocker analysis for Rx AGC section evaluation for Rx IC, QEC section evaluation using API and AD9528 Clock section evaluation using low level scripts
- Interacting with ADI technical team about Board Bring up issues and NEC technical team for status update and technical issues
- Schematic Capture for Clemente RF and Power supply section using cadence Allegro 16.5 and Layout Support with CAD team and review of layout
- Review of RF design specification document, Test report PPT preparation
- Tx and RX board Bring Up and evaluation of Tx and RX performance using Test models as per the 3GPP requirement
- Review of RF design specification document, Test report PPT preparation

13. Project Name: WCDMA Multi Carrier Power Amplifier design (MCPA)

Client: NSN
Organization: Wipro Technology
Software Platform: AWR Microwave Office, AppCAD
Team Size: 2
Duration: Jan’2012 to Aug’2012
Description: MCPA Sub-assembly’s task is amplifying TX-RF signal to 5W output power level at defined maximum DC power consumption, raw linearity and linearization. SPAC Sub-assembly provides coupled RF output signal to feedback chain (RF_FB) for linearization using pre-distorter. It includes three amplifier stages: Pre-driver, Driver and final Power stage. Final stage is a PA IC device that consists of driver and corresponding output transistors in a single package. Final stage is preceded by two stage amplification. The driver stage is running at +5.0 VDC drain supply. A linear regulator (5V) will be used for stable and accurate drain supply for the driver device. Final stage devices operate with +28VDC drain supply. The gate bias for the final stage is derived from 28V DC supply, through a network of resistor and EPOTs for tuning the gate bias and thereby adjusting the quiescent currents

Contribution:

- Involved in PA design and validation which Involves device selection, Simulation of the model and layout preparation
- Preparation of Line-up Calculation
- Preparation of derating analysis document
- Preparation of Implementation Specification,TestPlan and Test Specification document
- Schematic Capture and layout review
- System Level testing and Integration.

14. Project Name: Directional Of Arrival (DOA) for Building application

Organization: Honeywell Technology Solutions Lab Pvt. Ltd
Software Platform: AWR Microwave Office, APPCAD
Team Size: 2
Duration: Jan’2011 to May’2011
Description: Direction Of Arrival (DOA) is a System for estimation of accuracy of fire Intruder in Building application. It is a proto design for R&T which Contains 36Receiver chains to down convert signals from 902-928MHz to base band frequency of DC to 90MHz,the demodulated quadrature I&Q O/p’s are directly connected to 16-bit 64-Channel PMC66-16AI64SS Analog i/p card for simultaneous sampling at rates of up to 200KSPS with SW selectable Analog i/p ranges of +/-10V, +/-5V, +/-2.5V,0/+5V,or 0/+10V,the Antenna geometry is 6x6 3D array antenna from 3rd Party; the antenna design is matched to operate in frequency band of 902 to 928MHz

Contribution:

- Involved in discrete Rx design and validation which involves component selection, Line-up calculation Simulation of Components and Schematic capture
- Line up calculation for the system design
- Preparation of design document and Test report
- System Level testing and Integration
- Integration of Receiver Chain with PMC card

15. Project Name: RDR-4000 Product Maintenance and Improvements (PMI)

Organization: Honeywell Technology Solutions Lab Pvt Ltd
Software Platform &Design Tools: AWR Microwave office, Partminer, AeroPDM, Honeywell Obsolescence Management System, Pspice
Team Size: 2
Duration: Oct’2008 to Sep’2011
Description: The RDR-4000 Receiver-Transmitter is part of the Honeywell Aerospace Electronic Systems with Forward-Looking Wind shear Detection/Avoidance capability. The RDR-4000 Receiver-Transmitter is a lightweight, completely solid-state airborne unit which performs the transmitting, receiving, video processing, Antenna stabilization and system timing and monitoring, for the RDR-4000 Airborne Weather Radar System with Forward-Looking Wind shear Detection/Avoidance capability. The RDR-4000 provides Pulse transmission in the X-band frequency range for detection of Storm Conditions (or) Terrain mapping at ranges up to 320 nautical miles. It can service up to three independent display devices. The unit also provides predictive wind shear hazard-factor information and warning for microburst wind shear during take-off and approach

Contribution:

- Testing and fault rectification activity in sustenance team for Next Generation Weather Radar System (RDR-4000)
- Design Support for improving the Factory Yields, Obsolete Part Replacement, & Cost take out Opportunities
- Managing the Obsolete Part Database and Providing timely solutions for Factory

16. Project Name: Modulator Circuit for the RDR-4000 Transmitter Improvement

Organization: Honeywell Technology Solutions Lab Pvt Ltd
Software Platform &Design Tools: Spice,
Team Size: 1
Duration: Apr'2009 to Oct'2009

Description: Modulator Circuit will Modulate the FET Drain Voltage, Maximum Duty Cycle is 10% and Maximum Pulse width is 550uS, designed for 40Amps of peak Current and 4 Amps of Average Current used to turn ON and OFF the final Power Amplifier

Contribution:

- Driver Circuit IC and High Power MOSFET device model selection, Pspice Simulation of the Individual IC's and Complete Modulator Circuit.
- Schematic Capture, Pizza Board Development, Layout review and Validation of the design

17. Project Name: Multi diversity Receiver

Organization: Honeywell Technology Solutions Lab Pvt Ltd
Software Platform&& Design Tools: AWR Microwave Office
Team Size: 2
Duration: Oct'2007 to Feb'2008
Description: Main merits of this Rx is the overall Noise figure of the system is minimum (ie.,<3dB) and by having four Rx in parallel rugged to operate in multi path environment, Interference scenarios and improves the overall SNR of system. The Receiver section has four parallel receivers with IF of 2MHz output to ADC from RF in 1MHz bandwidth. The operation principle is similar to that of 1x4 MIMO Rx

Contribution:

- Individual Module design and testing, Link budget analysis
- Schematic Capture and layout review
- Involved in Integration of Rx module with ADC and Signal processing

18. Project Name: Wireless Kingfisher Module

Organization: Honeywell Technology Solutions Lab Pvt Ltd
Design tools: AWR Microwave Office, Line Calc, AppCAD
Team Size: 5
Duration: Mar'2008 to Sep'2008
Description: Wireless Kingfisher module provides point to point communication link between Wall modules, Dongle and is used for Automatic Control Solutions (ACS) /Honeywell Building application platform to transmit and control the room temperature, Fan control

Contribution:

- Involved in Radio board design and validation
- The Radio board design with and without LNA and PA
- Link budget analysis and Battery life analysis for the system design
- System Level testing and Integration
- Preparation of quality documents such as Test plan, Test case, Functional Test Report document and Precompliance test report
- Involved in Precompliance testing (PCT) and FCC Product Certification
- Carried Range trials and system level Interference testing

19. Project Name: RAPTOR FOR WIRELESS NETWORK SECURITY FOR INDUSTRIAL APPLICATION (WNSIA)

Organization: Honeywell Technology Solutions Lab Pvt Ltd
Design tools: AWR Microwave Office
Team Size: 2
Duration: April'2006 to Feb'2008
Description: WNSIA System architecture consists of Leaf nodes that are typically battery powered and Infrastructure nodes (I-nodes) that are line powered. The Leaf nodes either transmit data from Sensors (or) Receive data from the I-nodes(Actuators). The I-nodes take over task of relaying the data from Leaf nodes to the control centre (or) relaying the data from control centre to the other I-node. The wireless communication between Leaf nodes and the I-nodes are based on Frequency Hopping Spread Spectrum (FHSS) technology and designed keeping in mind requirement such as low power consumption at battery powered leaf nodes and increased reliability of the wireless communication. The leaf nodes wakeup according to their application duty cycle, transmit data using FHSS, Receive acknowledgement from I-nodes and go to sleep mode. Involved in both the versions **ie.,** R100 and R110 of design, the technology involved for the R100 is FHSS whereas for R110 is DSSS.

Contribution:
- Involved in Radio board design and validation
- The Radio board design with LNA , PA and also without PA
- Link budget analysis and Battery life analysis for the system design
- System Level testing and Integration
- Preparation of quality documents such as Test plan, Test case, functional Test Report document and Precompliance test report
- Involved in PCT and FCC Product Certification
- Involved in Range trials and system level Interference performance testing

20. Project Name: RECEIVER MODULE FOR TACTICAL AIR NAVIGATION SYSTEM (TACAN)

Client: INDIAN AIR FORCE
Organization: Hindustan Aeronautics Ltd (HAL), Hyderabad
Design Tools: Advanced Design System, EESOF, LINECALC, MATLAB,PSpice
Team Size: 3
Duration: Jan'2004 to Apr'2006
Description: TACAN system is used to guide the pilot by providing him the information about distance and bearing information. TACAN Receiver is Super heterodyne type capable of handling simultaneous signals with in the frequency range of 962-1213 MHz from various ground stations (interrogations) with excellent sensitivity of -90dBm and dynamic range of 90dBm (ie.,0 to -90 dBm) and the received signal will down converted in to IF frequency of 10.7 MHz and the IF output will detected and amplified using video amplifier and video information given to the signal processing for further encoding process. This project is newly developing for Su-30 aircraft communication
Contribution:
- Individual Rx Module design and testing
- Integration with Transmitter and signal processing
- System Level Simulation and Integration.
- Involved in Preparation of quality documents such as project plan document, Design document and Acceptance test plan for Receiver subsystem.
- EMI/EMC testing and integration of total system

21. Project Name: DESIGN && DEVELOPMENT OF GaASFET LOW NOISE AMPLIFIER AT Ku-BAND

Client/Organization: SLRDC, HAL, HYDERABAD
Design Tools: EESOF, AutoCAD
Instruments Used: Network Analyzer, Noise figure meter
Team Size: 1
Duration
Feb'2004 to June'2004
Description:
Low Noise Amplifiers are important to process weak Signals these receivers plays important role in deciding sensitivity and dynamic range of the system with increasing requirements of low noise figure, higher Gain LNA with better technology called GaAsFET are used, the designed LNA in this project has two stages:

i. Design of Low Noise figure stage
ii. Design of Maximum gain stage.

Design is carried out using EEsof touch stone RF software, once design is completed layout has been developed and validated for the performance, the testing results shows they are meeting the desired specifications.

Contribution:

- Design and development of two stages of LNA
- Testing of developed module using Noise figure meter and Network Analyser
- Understanding of Design concept and project document preparation.

Academic Project:

22. Project Name: IMPLEMENTATION OF FLOATING POINT ALGORITHMS USING VHDL

Client/Organization: TTM INDIA PRIVATE LTD, HYDERABAD
Development Tools: WARP Software, Modelsim, Mentor Graphic Synthesizer
Description: To support floating-point operations necessary for implementing Digital filters. Floating point operations are quite often included in the internal hardware. If no hardware is available for these operations compiler must be designed with package of floating point software subroutines. In this project we developed components like adder, multiplier, and delay components using floating point representation of numbers using VHDL.

Contribution:

- Design concept and Preparation of project documentation
- Major part coding implementation, Testing and Guidance to other team members

Designed Component Profile:

S.no	Module Name	Project	Technology
1.	L- Band 2-Way Power Divider	TACAN	Microstrip
2.	L- Band Low pass Filter	TACAN	Microstrip
3.	L- Band Directional Coupler	TACAN	Microstrip
4.	L- Band AGC System	TACAN	Microstrip
5.	L- Band Variable attenuator	TACAN	Microstrip
6.	Ku-Band Low Noise Amplifier	MMR	Microstrip
7.	X-band 1:5, 1:6, 1:7,1:10 Equal and Unequal Power Divider	Seeker	Microstrip
8.	1:2 CPWG Power Divider	Seeker	CPWG
9.	1:16 Power Combiner	Front End Receiver	Microstrip
10.	1 to 6GHz High Pass filter Design	WidebandTRM	Microstrip
11.	10W Power Amplifier design using CGH40120F	Seeker	Microstrip
12.	10W Power Amplifier design using CGH40006-P	Seeker	Microstrip
13.	S-band 10W Power Amplifier design using NPT1012 Transistor	Front End Receiver	Microstrip
14.	S-band Low Noise Amplifier (LNA) design using NE3210S01 Transistor	Front End Receiver	Microstrip
15.	10.8GHz Edge Coupled BPF Design	Seeker	Microstrip
16.	500MHz,1GHz and 2GHZ Lumped filter BPF design	Front End Receiver	Microstrip
17.	3-4GHz, 6-8GHz Hairpin BPF design	Fixed LO (RSA P&R)	Microstrip
18.	6-8GHz Edge Coupled BPF design	Fixed LO (RSA P&R)	Microstrip
19.	2.89-3.3GHz Band Pass Filter design	S-band RTS	Microstrip
20.	1:2,1:3,1:4, 1:6 and 1:8 500MHz to 6GHz Wideband Power dividers Design	PD-ELINT	Microstrip
21.	9.3-9.6GHz Hairpin Filter and Edge CPLD Filter design	AD RADAR	Microstrip

22.	5-6GHz High Power Directional Coupler , Rat Race CPLR and 1:2 PD design	DEJ Wideband Jammer	CPWG

Personal Profile:

Name	:	Badam PraveenKumar
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Nationality	:	Indian
Marital Status	:	Married
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- I hereby declare that all information provided above is true to the best of my knowledge. Finally, I request the concerned authorities to provide me a chance, so that I can prove the best of myself and serve your esteemed organization for which act of magnanimity, I shall ever be thankful and grateful to you

(**Badam PraveenKumar)**